

Shail Patel

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SUMMARY:

Results-driven Data Engineer with 5 years of experience in designing, building, and optimizing scalable data pipelines and architectures within Apache, AWS and Databricks. Expertise in efficient ETL processing and big data management. Proficient in SQL and NoSQL databases and skilled in Python and PySpark for data transformation and automation. Strong knowledge of data modeling, warehousing, and security best practices, ensuring compliance and high-performance analytics. Passionate about leveraging cloud services to drive real-time data processing with Kafka, business intelligence, and AI-driven insights.

CORE QUALITIES:

Programming & Scripting: Python (Pandas, PySpark), SQL

Databases & Storage: PostgreSQL, MySQL, NoSQL (MongoDB)

Data Integration & ETL: Apache Spark, AWS Glue, AWS Lambda, AWS Kinesis, Databricks

Data Warehousing & Modeling: Amazon RDS, Amazon Redshift, Star/Snowflake Schema

Monitoring & Performance Optimization: Query Performance Tuning, Spark Job Optimization

Big Data: Hadoop, Apache Kafka, Amazon EMR

Automation & DevOps: CI/CD Pipelines, MWAA

Data Security & Governance: Data Masking, GDPR & HIPAA Compliance Standards

Education:

Master of Science in Data Science | New Jersey Institute of Technology, NJ, USA | **May 2021**

Bachelor of Engineering in Computer Engineering | Dharmasinh Desai Institute of Technology, India | **April 2018**

Professional Experience:

UnitedHealth Group, USA | Data Engineer | Sept 2024 - Present

Responsibilities:

- Designed and implemented scalable ETL pipelines using Amazon EMR, PySpark, and Spark SQL, processing terabytes of structured and unstructured data from diverse sources into Amazon S3 for advanced analytics and reporting.
- Optimized Big Data workflows by leveraging Apache Spark on Amazon EMR and Databricks, reducing data processing time by 40% and enhancing system efficiency for real-time and batch analytics.
- Developed and maintained robust data models in Amazon Redshift and Amazon RDS, ensuring data integrity and enabling seamless integration with business intelligence and reporting tools like Amazon QuickSight.
- Automated data ingestion and transformation processes using Python, and Databricks Notebooks, incorporating data quality checks and validation frameworks to support high-quality governance and GDPR compliance.
- Architected and deployed high-performance, scalable data solutions across AWS, ensuring secure and efficient data integration between Amazon S3, Amazon Redshift, Amazon EMR, and visualization tools for real-time business insights.
- Collaborated with cross-functional teams, including data scientists and DevOps engineers, to develop CI/CD pipelines using AWS MWAA, and AWS IAM for automated, secure, and version-controlled data workflows.

Environment: AWS (Amazon Redshift, Amazon EMR, Amazon S3, Amazon RDS, AWS CloudWatch, Amazon Kinesis), Apache Spark, Spark SQL, Databricks, Hadoop, Kafka, Python, SQL, HIPAA Compliance.

JMSC, India | Data Engineer | Nov 2022 – May 2024

Responsibilities:

- Led a large-scale data migration project, transforming and loading high-volume datasets into Amazon Redshift and Amazon RDS, ensuring high data accuracy, lineage tracking, and performance optimization across distributed systems.
- Developed and maintained robust ETL/ELT pipelines using Amazon EMR, AWS Glue, and Amazon Redshift, enabling seamless data ingestion, transformation, and integration from multiple cloud and on-prem sources.
- Designed dynamic, real-time data visualizations using Apache Spark and Amazon QuickSight, allowing stakeholders to explore key business metrics and drive data-informed decisions.
- Optimized Amazon Redshift performance by implementing partitioning, clustering, and table sharding strategies, significantly improving query performance and reducing storage and processing costs.

- Created interactive dashboards and dynamic visualizations using PySpark, enabling real-time data exploration and actionable insights across business teams.
- Integrated and processed data from diverse sources including APIs, Amazon S3 and relational databases, ensuring reliable, high-throughput AWS Glue across platforms and domains.

Environment: AWS (Amazon Redshift, Amazon EMR, AWS Glue, Amazon S3, Amazon RDS), Apache Spark, Spark SQL, Kafka, Python.

Infosys (Client Goldman Sachs), USA | Data Engineer | July 2021 - Oct 2022

Responsibilities:

- Developed and optimized scalable ETL pipelines using Amazon MWAA (Apache Airflow), Amazon EMR, and Spark SQL to process large-scale financial datasets, enabling real-time data streaming and transformation with Kafka and Cloud Functions.
- Built and maintained secure and efficient data lakes and data warehouses on Amazon S3 and Amazon Redshift, leveraging Redshift Spectrum and Amazon Redshift Federation for optimized querying across structured and semi-structured sources.
- Automated data workflows and infrastructure provisioning using Terraform, Jenkins, and Git, improving deployment efficiency and integrating seamlessly with Tableau for near real-time analytics and business intelligence reporting.

Environment: AWS (Amazon Redshift, Amazon EMR, Amazon S3), Kafka, Spark, Spark SQL, Databricks, Hadoop, Python, SQL, Git, Airflow, Security & Compliance (GDPR).

Radix InfoTech Pvt. Ltd., India | Python Developer | Apr 2018 - Jul 2019

Responsibilities:

- Developed Python applications to automate business workflows and data extraction processes.
- Created custom Python scripts for data transformation, reducing manual data processing tasks by 50%.
- Integrated external APIs to retrieve real-time data and synchronize it with internal systems.

Environment: Python, SQL, MySQL.